Vinay Somala

+49 155 65746358 | vinay.doer@gmail.com | Hamburg, Germany
LinkedIn: linkedin.com/in/vinaysomala | **GitHub:** github.com/somalavinay879-18 | **Portfolio:** somalavinay.vercel.app

Data Science | LLMs & RAG | Machine Learning | Python | SQL | SAP S/4HANA MM | Supply Chain

SUMMARY

M.Sc. Applied Data Science & AI student at SRH University Hamburg, specialising in Supply Chain Analytics, with hands-on SAP S/4HANA MM training and a software engineering internship. I build end-to-end data pipelines, machine-learning models, LLM applications with retrieval-augmented generation and KPI dashboards with Python and SQL, and back them with automated tests and data quality checks. My SAP training covers the full Procure-to-Pay process, so I understand the business processes behind purchasing, delivery and inventory data. Seeking a working-student role or internship in data science, data analytics, SAP MM or supply chain analytics in Hamburg or remote.

EDUCATION

SRH University Hamburg — Oct 2025 – Sep 2027 (expected)
M.Sc. Applied Data Science & Artificial Intelligence — *Hamburg, Germany*

- **Specialisation:** Supply Chain Analytics
- **Relevant coursework:** Machine Learning, Artificial Intelligence, Statistical Modelling, Database Systems, Demand Forecasting, Inventory Management

Vedavyasa Degree College, Sri Venkateswara University — 2021 – 2024
B.Sc. Mathematics, Statistics & Computer Science — *India*

- **Core subjects:** Mathematics, Statistics, Computer Science, Programming, Database Management

EXPERIENCE

Karthikeyan Software Solutions Pvt. Ltd. — Jan 2023 – Jul 2023
Full Stack Engineer Intern — *India*

- Built responsive web components with HTML, CSS, JavaScript and PHP for client web applications.
- Worked with MySQL for data storage and retrieval behind application features.
- Supported debugging, testing and Git-based code changes within the development team.

PROJECTS

LexiSyncAI – Evidence-Grounded Legal Q&A (LLM & RAG) — GitHub
Python, Ollama (Qwen 2.5 7B), sentence-transformers, BM25, PyMuPDF, Tesseract OCR, Streamlit

- Built a local RAG system that answers questions about legal PDFs and cites the exact page and clause for every claim, highlighting the cited region on the page.
- Parsed PDFs with two-column reading order, tables and OCR for scanned pages (96.2% character accuracy), chunked by article and clause.
- Combined BM25 and embedding search with rank fusion and a cross-encoder reranker; paragraph-level recall@1 rose from 0.53 (dense) to 0.72.
- Added claim verification and abstention: correctness rose from 56% (LLM alone) to 78%, and all unanswerable questions were refused instead of answered with invented facts.
- Validated on an unseen contract with held-out questions: 7/8 correct and 100% citation precision; 29 tests run in GitHub Actions.

P2P Process Intelligence – SAP Purchase-to-Pay Process Mining — GitHub
SQL, DuckDB, Python, scikit-learn, Streamlit, pytest, GitHub Actions

- Process-mined a real SAP purchase-to-pay event log (BPI Challenge 2019: 1.46 M events, 251,734 PO items, €1.0 B) in DuckDB SQL: directly-follows graph, variants and step durations.
- Found that payment takes a median 78 days and the wait is in finance (42 days from invoice to payment vs 10 days to goods receipt); payment blocks hit 22% of items and add 14 days.
- Checked every item against its own 3-way / 2-way matching rule: 654 items worth €1.75 M were paid before the goods receipt.
- Built a payment-block early warning with a time-based split and leakage-safe vendor history: the riskiest 10% of new orders contain a block in 42% of cases vs 23% at random.
  Live dashboard: Open the live Streamlit dashboard

Olist Retail Analytics – Sales, Delivery & Demand **GitHub**
Python, SQL, DuckDB, scikit-learn, Streamlit

- Built an end-to-end pipeline on 99,441 real e-commerce orders (2016–2018): raw CSVs loaded into DuckDB, cleaned in SQL staging and modelled as a star schema with 2 fact and 4 dimension tables.
- Wrote 20 automated data quality checks (duplicate keys, orphan foreign keys, impossible dates, revenue reconciliation) and pytest tests on a synthetic dataset, running in GitHub Actions on every push.
- Found that late orders average a 2.27 review score vs 4.29 for on-time orders; the late rate peaked at 19.0% in March 2018 and is 12.1% in Rio de Janeiro vs 4.5% in São Paulo.
- Forecast weekly demand for the 10 largest categories, compared gradient boosting with two baselines (best WAPE 22.6%) and turned the forecast error into safety stock and reorder points.
  Live dashboard: **Open the live Streamlit dashboard**

Cryptocurrency Price Direction Prediction **GitHub**
Python, scikit-learn, XGBoost, pandas, pytest, GitHub Actions

- Predicted the next 15-minute price direction for five crypto pairs from 851,505 candles, using past-only returns, lags and rolling volatility.
- Evaluated on a strict time-based split (last 365 days held out): Random Forest and XGBoost reached 52.5% accuracy (ROC-AUC 0.536) vs 52.1% for logistic regression and 50.2% for a majority-class baseline.
- Refactored the notebook into a Python package with 12 pytest tests, including a check that future prices never change earlier features, run in GitHub Actions.

Demand Forecasting & Inventory Simulation
AnyLogistix, Supply Chain Analytics

- Modelled an end-to-end supply chain (customers, factories, distribution centres, demand) and compared inventory and distribution scenarios.

SAP TRAINING

ERP4Students, Technische Hochschule Brandenburg Summer 2026 – Completed
SAP MM – Sourcing & Procurement with SAP S/4HANA *Online*

- Completed the Procure-to-Pay process in SAP Fiori and SAP GUI: purchase requisition, RFQ and quotation, purchase order, goods receipt and invoice verification.
- Created and maintained material and vendor master records, purchasing info records and source lists, and worked with the enterprise structure.

ERP4Students, Technische Hochschule Brandenburg Winter 2026/27 – In progress
SAP MM – Inventory Management & Invoice Verification *Online*

- Studying inventory management, goods movements, stock transfers and invoice verification in SAP S/4HANA.

SKILLS

Programming & Analysis: Python, SQL, Pandas, NumPy, Statistical Modelling, Exploratory Data Analysis, Excel (PivotTables, XLOOKUP)
Machine Learning: scikit-learn, XGBoost, Random Forest, Logistic Regression, Feature Engineering, Time-Series Forecasting, Model Evaluation
LLMs & GenAI: Retrieval-augmented generation (RAG), BM25 + embedding hybrid search, cross-encoder reranking, Ollama, prompt and output design, LLM evaluation
Data Engineering: ETL Pipelines, DuckDB, MySQL, SQLite, MongoDB Atlas, Data Quality Checks, pytest, GitHub Actions
BI & Visualisation: Power BI, Tableau, Streamlit, Plotly, Matplotlib, KPI Dashboards
SAP & Supply Chain: SAP S/4HANA MM, SAP Fiori, SAP GUI, Procure-to-Pay, Process Mining, Material and Vendor Master Data, Demand Forecasting, Safety Stock, Reorder Points, AnyLogistix
Tools: Git, GitHub, Docker, Flask, REST APIs, Jupyter, VS Code, Linux, AWS (basics)
Languages: English (professional), German (A2, learning), Telugu (native), Hindi (professional)

CERTIFICATIONS

- **SAP:** Introducing Joule Studio, completion badge (2026); Joule Studio certification exam, Dec 2026
- **AWS:** AWS Academy Graduate – Machine Learning Foundations and Cloud Foundations (2022)
- **DataCamp:** Data Analyst with Python track; SQL Certification track